Primal and dual bounds

Outline

Why a relaxation?

Primal and dual bounds

Primal bounds: heuristic solutions

The LP relaxation

Rounding the solution of the LP relaxation

Upper bounds on the variables

Dual bounds

    Summary of the notation

Typical mistakes

In summary

Learning objectives (1/2)

At the end of this chapter you will be able to:

- obtain a primal bound (a lower bound) on the optimal value from a feasible solution, also with a simple heuristic algorithm;
- build the LP relaxation of an ILP problem and use its optimal value as a dual bound (an upper bound), rounded down when it is valid;
- compute the optimality gap from a primal bound and a dual bound;

Learning objectives (2/2)

At the end of this chapter you will be able to:

- explain why rounding an optimal LP solution gives a feasible solution at best;
- derive upper bounds on the variables from the constraints, and say when they help;
- obtain a dual bound from non-negative multipliers of the constraints, and recognize the dual problem as the problem of the best multipliers.

Prerequisites

- From this module, the chapter «Integer linear programming»: ILP problems, feasible and optimal solutions, complete enumeration, and the car example
- From the module *Linear Programming*, Chapter 9 «Duality theory»: the dual problem, weak duality and strong duality
- From the same module, Chapter 6 «Simplex algorithm»: an LP problem that has an optimal solution has an optimal vertex

Where we were

- In the previous chapter we saw that complete enumeration is not an option
- We need bounds on the optimal value from both sides
- We now obtain them:
  - **primal bounds** from feasible solutions
  - **dual bounds** from relaxations

Why a relaxation?

Introduction: Why a relaxation?

- A feasible solution tells us how good the optimum is *at least*
- To certify optimality we also need to know how good it is *at most*
- We see why this second bound needs a different idea: relaxing the problem

A feasible solution gives only one side

- A feasible solution proves that the optimal value is **at least** its value
- To prove that no feasible solution has value larger than a given number, we cannot look at the solutions one by one
- We need a statement that holds for all the feasible solutions at once

The idea: relax the problem

- We **relax** the problem: we replace it with an easier problem whose feasible region contains the original one
- A maximization over a larger region cannot give a smaller value
- So the optimal value of the relaxed problem is an **upper bound**

The easier problem we choose is the one we know how to solve:
a **linear programming** problem.

Summary

- A feasible solution gives a bound from below only
- A bound from above must hold for all the feasible solutions at once
- A relaxation has a larger feasible region, hence a larger (or equal) optimal value
- The relaxation we use is a linear programming problem

Primal and dual bounds

Introduction: Primal and dual bounds

- We define the two kinds of bounds on the optimal value: **primal** and **dual**
- We measure how far apart they are with the **optimality gap**
- We define a **relaxation**, the general source of dual bounds
- These notions are used in every chapter of the module

Definition 1: primal bound, dual bound and optimality gap

Consider an ILP problem in canonical maximization form, with feasible region F and optimal value $z(\mathrm{ILP})$.

- *Given a feasible solution $\bar{\boldsymbol{x}} \in F$, its value $\mathrm{LB}(\bar{\boldsymbol{x}}) = \sum_{j=1}^{n} c_j \bar{x}_j$ is a **primal bound** on $z(\mathrm{ILP})$.*
- *A value UB such that $\sum_{j=1}^{n} c_j x_j \leq \mathrm{UB}$ for every $\boldsymbol{x} \in F$ is a **dual bound** on $z(\mathrm{ILP})$.*
- *Given a primal bound $\mathrm{LB} \neq 0$ and a dual bound UB, the **optimality gap** is*

$$\textit{optimality gap} = \frac{\mathrm{UB} - \mathrm{LB}}{|\mathrm{LB}|}.$$

Lower bounds and upper bounds

- In a maximization problem:
  - the primal bounds are **lower bounds**
  - the dual bounds are **upper bounds**
- A primal bound comes from **one** feasible solution
- A dual bound is a statement on all the feasible solutions at once

Certified optimality

For every feasible solution $\bar{\boldsymbol{x}}$ and every dual bound UB:

$$\mathrm{LB}(\bar{\boldsymbol{x}}) \;\leq\; z(\mathrm{ILP}) \;\leq\; \mathrm{UB}.$$

A feasible solution is **certified optimal**
when the optimality gap is zero.

Where do dual bounds come from?
From problems with a **larger** feasible region:
the relaxations.

Definition 2: relaxation

Given an ILP problem $\max\{\boldsymbol{c}'\boldsymbol{x} : \boldsymbol{x} \in F\}$, *the problem* $\max\{\boldsymbol{c}'\boldsymbol{x} : \boldsymbol{x} \in R\}$ *is a* ***relaxation*** *of the ILP problem if* $F \subseteq R$. *We denote its optimal value by* $z(R)$.

- A maximization over a larger region cannot give a smaller value: the optimal value of a relaxation is a dual bound,

$$z(\mathrm{ILP}) \leq z(R).$$

Summary

- **Primal bound** (lower bound): the value $\mathrm{LB}(\bar{\boldsymbol{x}})$ of a feasible solution
- **Dual bound** (upper bound): a value UB that no feasible solution exceeds
- **Optimality gap**: $\frac{\mathrm{UB}-\mathrm{LB}}{|\mathrm{LB}|}$; a solution is certified optimal when the gap is zero
- A **relaxation** has a feasible region $R \supseteq F$, and $z(\mathrm{ILP}) \leq z(R)$

Primal bounds: heuristic solutions

Introduction: Primal bounds: heuristic solutions

- Primal bounds come from feasible solutions: the better the solution, the better the bound
- We define **heuristic algorithms**, which find feasible solutions quickly
- We try two simple heuristics on the car example, and compare their bounds

Definition 3: heuristic algorithm

*A **heuristic algorithm** is a method that finds feasible solutions quickly, with no guarantee of optimality. The value of every solution it finds is a primal bound.*

A good feasible solution is a good primal bound.

Example: primal bound

- Consider the feasible solution $(2, 2)$ of the car example
- Its value is

$$\mathrm{LB} = 3 \cdot 2 + 2 \cdot 2 = 10$$

- Hence

$$z(\mathrm{ILP}) \geq 10$$

To build integer solutions from the constraints
we need to round numbers down:
the **integer part** of a number.

Definition 4: of integer part of a number

*Given $y \in \mathbb{R}$, the **integer part** of y is*

$$\lfloor y \rfloor = \max\{q \in \mathbb{Z} : q \leq y\}.$$

Example: two heuristic algorithms for the car example (1/3)

- The **one-variable heuristic** sets a single variable to its maximum value, with the other one equal to 0, and keeps the better of the two solutions

1. Only x_1 ($x_2 = 0$):

$$\bar{x}_1 = \min\left\{\left\lfloor\frac{45}{10}\right\rfloor, \left\lfloor\frac{49}{4}\right\rfloor\right\} = \min\{4, 12\} = 4,$$

so $\bar{\boldsymbol{x}} = (4, 0)$ and $\mathrm{LB} = 12$

2. Only x_2 ($x_1 = 0$):

$$\bar{x}_2 = \min\left\{\left\lfloor\frac{45}{4}\right\rfloor, \left\lfloor\frac{49}{14}\right\rfloor\right\} = \min\{11, 3\} = 3,$$

so $\bar{\boldsymbol{x}} = (0, 3)$ and $\mathrm{LB} = 6$

Example: two heuristic algorithms for the car example (2/3)

- The **two-step heuristic** sets first one variable to its maximum value, and then the other one to its maximum value with the **residual right-hand sides**; it keeps the better of the two solutions

1. x_1 first: $\bar{x}_1 = 4$ uses 40 and 16, leaving 5 and 33. Then

$$\bar{x}_2 = \min\left\{\left\lfloor\frac{5}{4}\right\rfloor, \left\lfloor\frac{33}{14}\right\rfloor\right\} = \min\{1, 2\} = 1,$$

   so $\bar{\boldsymbol{x}} = (4, 1)$ and $\mathrm{LB} = 14$

2. x_2 first: $\bar{x}_2 = 3$ uses 12 and 42, leaving 33 and 7. Then

$$\bar{x}_1 = \min\left\{\left\lfloor\frac{33}{10}\right\rfloor, \left\lfloor\frac{7}{4}\right\rfloor\right\} = \min\{3, 1\} = 1,$$

   so $\bar{\boldsymbol{x}} = (1, 3)$ and $\mathrm{LB} = 9$

Example: two heuristic algorithms for the car example (3/3)

- The four solutions found by the two heuristics:

Heuristic	Order	$\bar{x}_1$	$\bar{x}_2$	LB
one-variable	x_1	4	0	12
one-variable	x_2	0	3	6
two-step	x_1 first	4	1	14
two-step	x_2 first	1	3	9

Quality versus effort

- The two-step heuristic gives a better primal bound, but it needs more computation
- This trade-off between the quality of the solution and the computational effort is typical of heuristic algorithms
- Here the best primal bound is $\mathrm{LB} = 14$, in $(4, 1)$
- A primal bound alone is not a certificate of optimality: a **dual bound** is needed

Summary

- A **heuristic algorithm** finds feasible solutions quickly, without a guarantee of optimality
- Every solution it finds gives a primal bound: a good solution is a good bound
- Car example: the one-variable heuristic gives $\mathrm{LB} = 12$, the two-step heuristic gives $\mathrm{LB} = 14$ in $(4, 1)$
- Better solutions usually cost more computation

The LP relaxation

Introduction: The LP relaxation

- We drop the integrality constraints and obtain an LP problem: the **LP relaxation**
- Its optimal value is a dual bound, and with integer data it can be rounded down
- On the car example we get the first upper bound, and the first optimality gap

Definition 5: LP relaxation

*The **linear programming (LP) relaxation** of an ILP problem in canonical maximization form is the LP problem obtained by replacing the integrality constraints $x_j \in \mathbb{Z}_{\geq 0}$ with the non-negativity constraints $x_j \geq 0$, for all $j \in \{1, 2, \ldots, n\}$:*

$$\max \sum_{j=1}^{n} c_j \, x_j$$

$$\text{subject to} \quad \sum_{j=1}^{n} a_{ij} \, x_j \leq b_i, \quad \forall i \in \{1, 2, \ldots, m\},$$

$$x_j \geq 0, \quad \forall j \in \{1, 2, \ldots, n\}.$$

Notation

- We denote by $z(\mathrm{LP})$ the **optimal value** of the LP relaxation
- We denote by $\tilde{\boldsymbol{x}}(\mathrm{LP})$ an **optimal solution** of the LP relaxation
- We always write the argument, because in the same example $\tilde{\boldsymbol{x}}(\mathrm{ILP})$ and $\tilde{\boldsymbol{x}}(\mathrm{LP})$ are different points

Proposition 1: upper bound from the LP relaxation

Every feasible solution of the ILP problem is feasible for its LP relaxation, hence

$$z(\mathrm{ILP}) \leq z(\mathrm{LP}).$$

Moreover, if an optimal solution of the LP relaxation is integer, then it is an optimal solution of the ILP problem.

Proof

- The feasible region F of the ILP problem is contained in the polyhedron of the LP relaxation, and the objective function is the same
- The maximum over a larger set is not smaller than the maximum over a subset:

$$z(\mathrm{ILP}) \leq z(\mathrm{LP})$$

- If $\tilde{\boldsymbol{x}}(\mathrm{LP})$ is integer, it belongs to F and has value $z(\mathrm{LP}) \geq z(\mathrm{ILP})$: it is optimal for the ILP problem

□

The value of a feasible solution can be an integer,
and it is at most $z(\mathrm{LP})$:
then it is at most the largest integer not exceeding $z(\mathrm{LP})$.

Observation 1: rounding the upper bound

If all the variables are integer and all the costs c_j are integer, then

$$z(\mathrm{ILP}) \leq \lfloor z(\mathrm{LP}) \rfloor.$$

In all the other cases the bound $\lfloor z(\mathrm{LP}) \rfloor$ is not valid.

When can we round the bound?

- In this module all the ILP problems have integer data, so the bound $z(\mathrm{ILP}) \leq \lfloor z(\mathrm{LP}) \rfloor$ always holds
- In a MIP problem with a continuous variable, the value of a feasible solution is not necessarily an integer
- Then the bound **cannot be rounded**: the dual bound is $z(\mathrm{LP})$

Example: the LP relaxation of the car example (1/3)

- The LP relaxation of the car example has the optimal solution

$$\tilde{x}_1 = \frac{7}{2}, \quad \tilde{x}_2 = \frac{5}{2}, \qquad z(\mathrm{LP}) = \frac{31}{2},$$

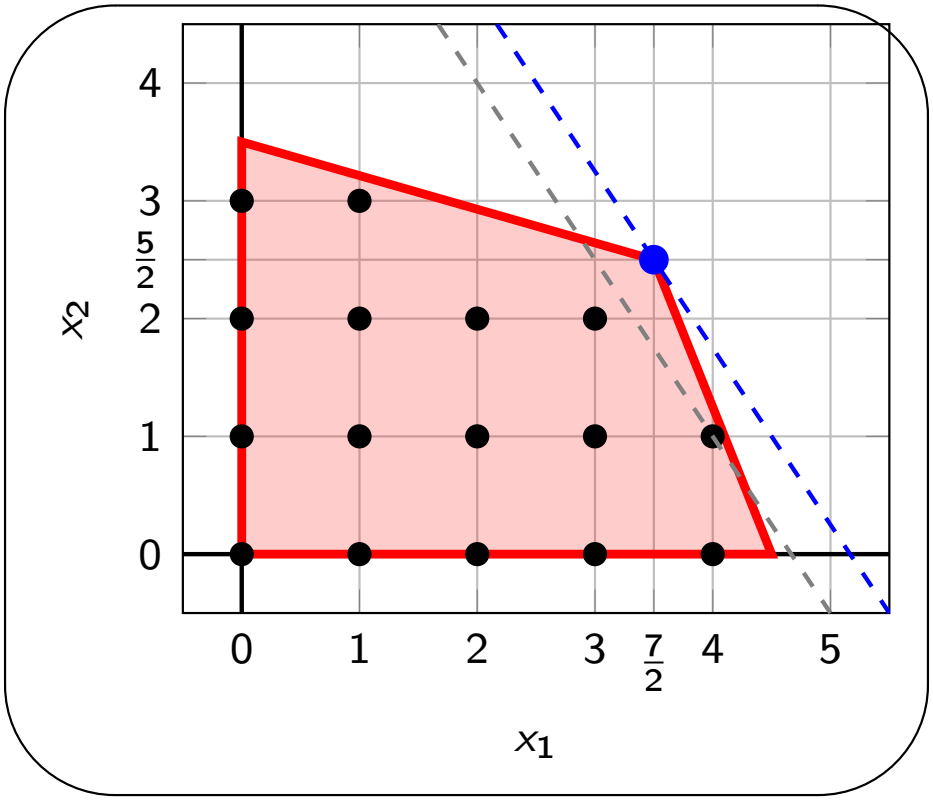
  and it is fractional: it is not feasible for the ILP problem

- Since the costs are integer, we have the upper bound $\lfloor z(\mathrm{LP}) \rfloor = 15$

- Together with the primal bound 14 of the feasible solution $(4, 1)$ found by the two-step heuristic:

$$14 \;\leq\; z(\mathrm{ILP}) \;\leq\; 15$$

Example: the LP relaxation of the car example (2/3)



The feasible region of the LP relaxation (red), its 16 integer points (black) and the optimal solution $(\frac{7}{2}, \frac{5}{2})$ (blue), with the objective line $3\,x_1 + 2\,x_2 = \frac{31}{2}$ through it (dashed). The gray dashed line is the objective line of the optimal ILP solution.

Example: the LP relaxation of the car example (3/3)

- The optimal value is 14 or 15
- With $\mathrm{UB} = 15$ and $\mathrm{LB} = 14$ the optimality gap is

$$\frac{\mathrm{UB} - \mathrm{LB}}{|\mathrm{LB}|} = \frac{15 - 14}{14} = \frac{1}{14} \approx 7.1\%$$

- The solution $(4, 1)$ is **not yet certified optimal**

Summary

- The **LP relaxation** replaces $x_j \in \mathbb{Z}_{\geq 0}$ with $x_j \geq 0$: it is an LP problem
- $z(\mathrm{ILP}) \leq z(\mathrm{LP})$, and an integer optimal solution of the LP relaxation is optimal for the ILP problem
- With integer variables and integer costs: $z(\mathrm{ILP}) \leq \lfloor z(\mathrm{LP}) \rfloor$; not in a MIP problem
- Car example: $z(\mathrm{LP}) = \frac{31}{2}$, so $14 \leq z(\mathrm{ILP}) \leq 15$, with a gap of about 7.1%

Rounding the solution of the LP relaxation

Introduction: Rounding the solution of the LP relaxation

- The optimal solution of the LP relaxation is often fractional: a natural idea is to round it
- We see three examples: rounding fails to give the optimum, rounding gives the optimum without proving it, rounding is far from the optimum
- The lesson: the **value** of the LP relaxation is useful, its **solution** is only a starting point

Rounding gives feasible solutions, not optima

Feasible ILP solutions can typically be obtained
by rounding an optimal solution of the LP relaxation.

But no rounding technique guarantees
to find an optimal ILP solution.

Example: rounding in the car example

- The optimal solution of the LP relaxation is $\tilde{\boldsymbol{x}}(\mathrm{LP}) = (\frac{7}{2}, \frac{5}{2})$
- Rounding to the nearest integer gives $(4, 3)$, which is infeasible:

$$10 \cdot 4 + 4 \cdot 3 = 52 > 45$$

- Rounding down both components gives the feasible solution $\bar{\boldsymbol{x}} = (3, 2)$, with $\mathrm{LB} = 3 \cdot 3 + 2 \cdot 2 = 13$
- The optimal ILP solution is $\tilde{\boldsymbol{x}}(\mathrm{ILP}) = (4, 1)$, with $z(\mathrm{ILP}) = 14$: rounding does not give the optimum, and the bounds are $13 \leq z(\mathrm{ILP}) \leq 15$

Example: rounding works, but the bound is not tight (1/3)

- Consider the following ILP problem:

$$\begin{array}{rrcrcr} \max & 2\,x_1 & + & 4\,x_2 & & \\ \text{subject to} & -\,x_1 & + & x_2 & \leq & 2 \\ & 8\,x_1 & + & 2\,x_2 & \leq & 19 \\ & x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0} \end{array}$$

- Its LP relaxation has $x_1, x_2 \geq 0$ instead of the integrality constraints

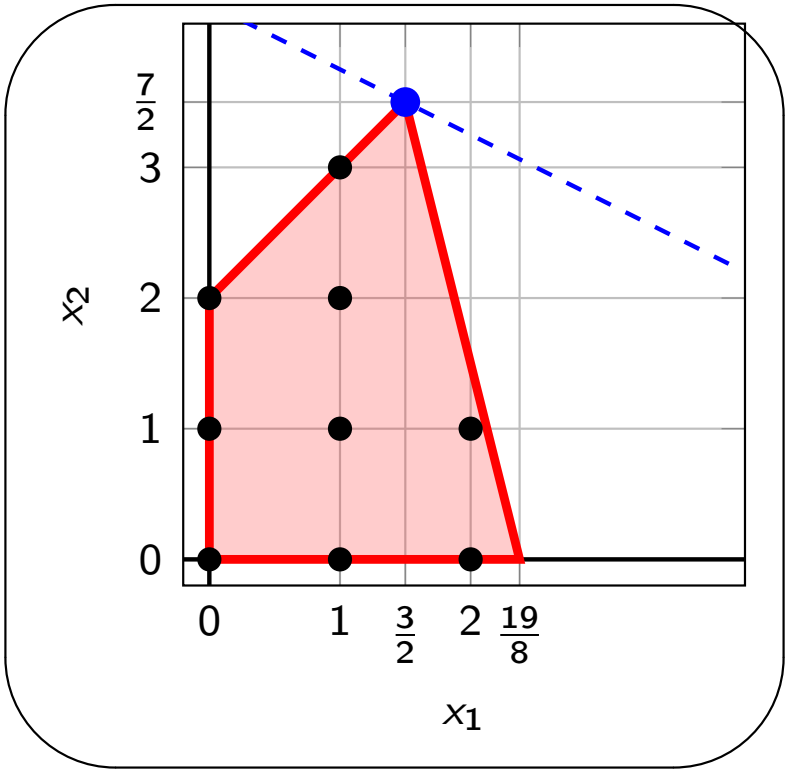
Example: rounding works, but the bound is not tight (2/3)

- The optimal solution of the LP relaxation is $\tilde{\boldsymbol{x}}(\mathrm{LP}) = (\frac{3}{2}, \frac{7}{2})$, with $z(\mathrm{LP}) = 17$ and $\lfloor z(\mathrm{LP}) \rfloor = 17$, so $z(\mathrm{ILP}) \leq 17$
- Rounding down both components we get the feasible solution

$$\bar{\boldsymbol{x}} = \left(\left\lfloor \frac{3}{2} \right\rfloor, \left\lfloor \frac{7}{2} \right\rfloor \right) = (1, 3), \qquad \mathrm{LB} = 2 \cdot 1 + 4 \cdot 3 = 14$$

- The optimal ILP solution is $\tilde{\boldsymbol{x}}(\mathrm{ILP}) = (1, 3)$, with $z(\mathrm{ILP}) = 14$: the rounded solution is optimal
- But the bounds $14 \leq z(\mathrm{ILP}) \leq 17$ **do not prove it**

Example: rounding works, but the bound is not tight (3/3)



The feasible region of the LP relaxation (red), the integer points (black) and the optimal solution of the LP relaxation (blue), with the objective line $2\,x_1 + 4\,x_2 = 17$ (dashed).

Example: rounding fails (1/3)

- Consider the following ILP problem:

$$\begin{array}{rrcrcr} \max & 25\,x_1 & + & 16\,x_2 & & \\ \text{subject to} & 50\,x_1 & + & 31\,x_2 & \leq & 250 \\ & -3\,x_1 & + & 2\,x_2 & \leq & 4 \\ & x_1, & & x_2 & \in & \mathbb{Z}_{\geq 0} \end{array}$$

- The optimal solution of the LP relaxation is

$$\tilde{\boldsymbol{x}}(\mathrm{LP}) = \left(\frac{376}{193}, \frac{950}{193}\right) \approx (1.94, 4.92),$$

$$z(\mathrm{LP}) = \frac{24600}{193} \approx 127.46$$

- Hence $z(\mathrm{ILP}) \leq \lfloor z(\mathrm{LP}) \rfloor = 127$

Example: rounding fails (2/3)

- Rounding to the nearest integer gives $(2, 5)$, which is infeasible:

$$50 \cdot 2 + 31 \cdot 5 = 255 > 250$$

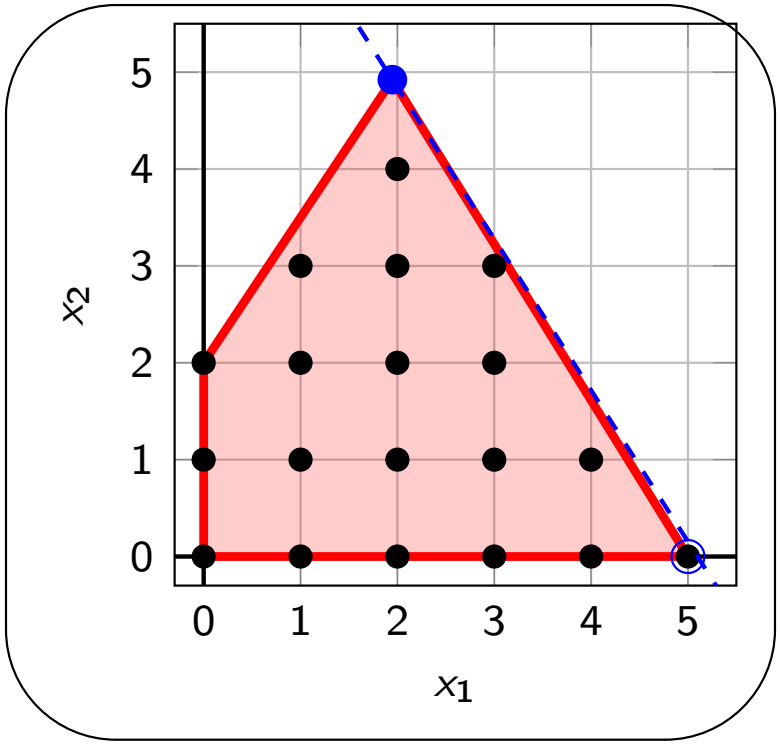
- Rounding $\tilde{x}_1$ up and $\tilde{x}_2$ down gives the feasible solution $\bar{\boldsymbol{x}} = (2, 4)$, with

$$\mathrm{LB} = 25 \cdot 2 + 16 \cdot 4 = 114$$

- The optimal ILP solution is $\tilde{\boldsymbol{x}}(\mathrm{ILP}) = (5, 0)$, with $z(\mathrm{ILP}) = 125$

- It is far from $\tilde{\boldsymbol{x}}(\mathrm{LP})$, and **no rounding** of $\tilde{\boldsymbol{x}}(\mathrm{LP})$ finds it

Example: rounding fails (3/3)



The integer points (black) are far from the optimal solution $(\frac{376}{193}, \frac{950}{193})$ of the LP relaxation (blue). The optimal ILP solution is $(5, 0)$ (blue circle). The dashed line is the objective line $25\,x_1 + 16\,x_2 = \frac{24600}{193}$.

Far solutions, good values

Optimal ILP solutions can be quite far
from optimal solutions of the LP relaxation.

The **value** of the LP relaxation, instead,
is often a good bound: here

$$125 \leq z(\mathrm{LP}) \approx 127.46.$$

Summary

- Rounding an optimal LP solution can give an **infeasible** point: $(4, 3)$ in the car example, $(2, 5)$ in the third example
- Rounding down can give a feasible solution, but not an optimal one: $(3, 2)$ in the car example
- Even when the rounded solution is optimal, the bounds may not prove it: $14 \leq z(\mathrm{ILP}) \leq 17$
- The optimal ILP solution can be far from the LP one, while $z(\mathrm{LP})$ stays a good bound

Upper bounds on the variables

Introduction: Upper bounds on the variables

- From a single constraint we can derive an upper bound on each variable
- These bounds hold for every integer point, so they can be added to the model
- We see when they improve the LP relaxation, and when they do not

Proposition 2: upper bounds on the variables

Let $\sum_{j=1}^{n} a_{ij}\, x_j \leq b_i$ *be a constraint with* $a_{ij} \geq 0$ *for all* j. *Then every feasible solution of the ILP problem satisfies*

$$x_j \leq \left\lfloor \frac{b_i}{a_{ij}} \right\rfloor, \qquad \forall j \text{ such that } a_{ij} > 0.$$

Proof

- All the variables are non-negative and all the coefficients are non-negative, hence

$$a_{ij}\, x_j \leq \sum_{k=1}^{n} a_{ik}\, x_k \leq b_i$$

- So

$$x_j \leq \frac{b_i}{a_{ij}}$$

- The variable is integer, so it does not exceed the integer part of $\frac{b_i}{a_{ij}}$:

$$x_j \leq \left\lfloor \frac{b_i}{a_{ij}} \right\rfloor$$



When do the bounds help?

- If several constraints give a bound on x_j, we keep the **smallest**
- The bounds cut off parts of the polyhedron of the LP relaxation, and no integer point

The upper bounds on the variables improve the LP relaxation only if they **cut off its optimal solution**.

Example: upper bounds on the variables of the car example (1/3)

- Both constraints of the car example have non-negative coefficients:

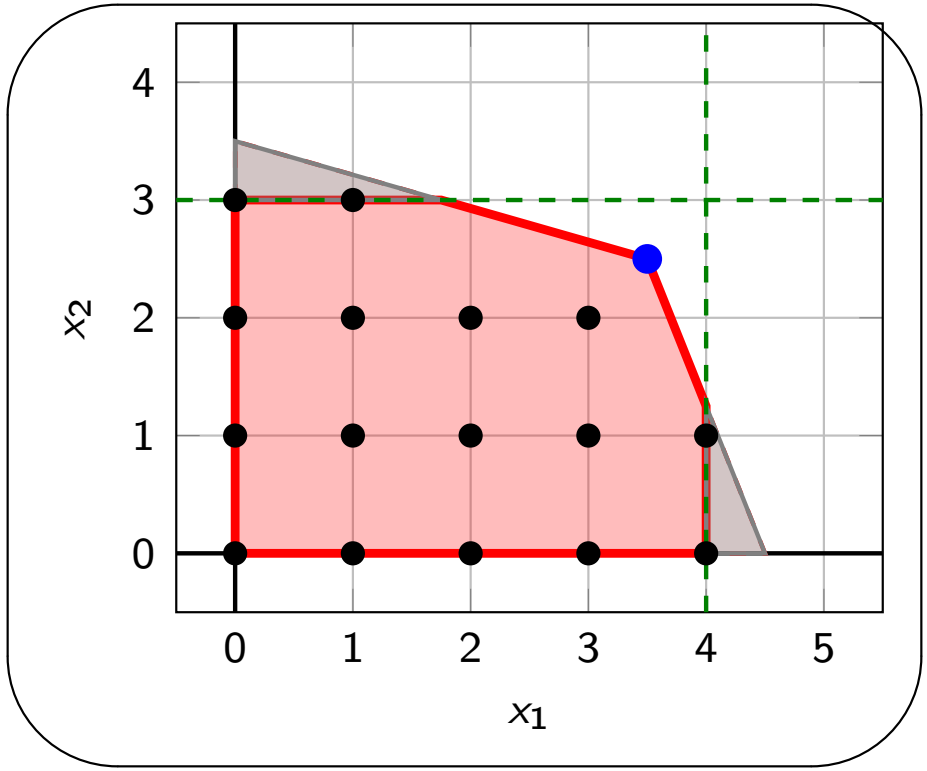
$$x_1 \leq \min\left\{\left\lfloor \frac{45}{10} \right\rfloor, \left\lfloor \frac{49}{4} \right\rfloor\right\} = \min\{4, 12\} = 4,$$

$$x_2 \leq \min\left\{\left\lfloor \frac{45}{4} \right\rfloor, \left\lfloor \frac{49}{14} \right\rfloor\right\} = \min\{11, 3\} = 3.$$

- The inequalities $x_1 \leq 4$ and $x_2 \leq 3$ are satisfied by all the integer points of the car example, and they can be added to the model

- They cut off two regions of the polyhedron, and no integer point

Example: upper bounds on the variables of the car example (2/3)



The variable upper bounds $x_1 \leq 4$ and $x_2 \leq 3$ (green) cut off the two gray regions of the LP relaxation (red) and no integer point. The optimal solution of the LP relaxation (blue) is not cut off.

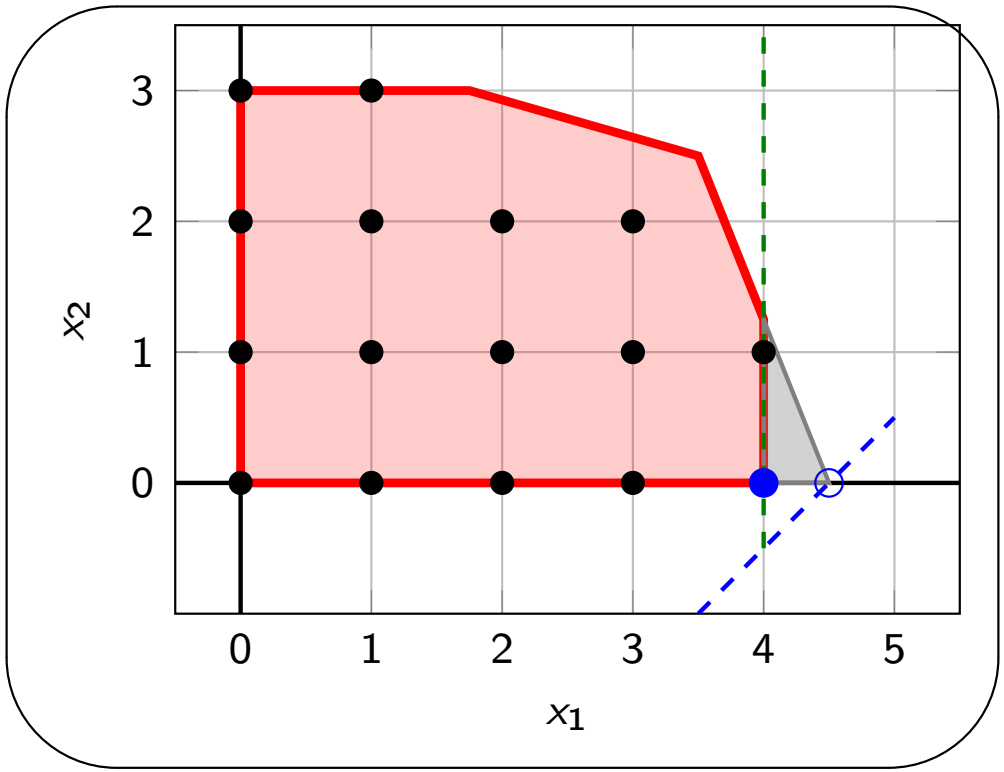
Example: upper bounds on the variables of the car example (3/3)

- The optimal solution of the LP relaxation, $(\frac{7}{2}, \frac{5}{2})$, satisfies both inequalities
- It is not cut off, so the optimal value of the LP relaxation does not change
- The upper bound is still 15

Example: upper bounds that cut off the LP optimum (1/3)

- Consider the objective function $x_1 - x_2$ on the constraints of the car example
- Its optimal ILP solution is $(4, 0)$, with $z(\mathrm{ILP}) = 4$
- Before the bounds, the optimal solution of the LP relaxation is $(\frac{9}{2}, 0)$, with $z(\mathrm{LP}) = \frac{9}{2}$
- After $x_1 \leq 4$ it is $(4, 0)$, with $z(\mathrm{LP}) = 4$, which is integer and therefore optimal

Example: upper bounds that cut off the LP optimum (2/3)



The objective $x_1 - x_2$: the variable upper bound $x_1 \leq 4$ (green) cuts off the gray triangle and the optimal solution $(\frac{9}{2}, 0)$ of the LP relaxation (blue circle). The new optimal solution $(4, 0)$ (blue) is integer. The dashed line is $x_1 - x_2 = \frac{9}{2}$.

Example: upper bounds that cut off the LP optimum (3/3)

- The value of the LP relaxation improves from $\frac{9}{2}$ to 4
- The rounded bound $\lfloor \frac{9}{2} \rfloor = 4$ was already 4

A better formulation does not always improve the rounded bound.

Summary

- A constraint with non-negative coefficients gives $x_j \leq \lfloor b_i / a_{ij} \rfloor$; with several constraints we keep the smallest bound
- The bounds cut off parts of the polyhedron and no integer point
- They improve the LP relaxation only if they cut off its optimal solution: not in the car example (still 15), yes with the objective $x_1 - x_2$
- A better value of the LP relaxation does not always improve the rounded bound

Dual bounds

Introduction: Dual bounds

- The LP relaxation gives a dual bound, but we have to solve it
- We see another way: combining the constraints, and reading off an upper bound
- The best multipliers solve the **dual problem**, and give exactly the value of the LP relaxation
- A dual feasible solution is a **certificate**: anyone can check it

Combining the constraints

- A dual bound can also be obtained without solving the LP relaxation
- We combine the constraints with **non-negative multipliers** π_i, one for each constraint
- The best multipliers are the solution of an LP problem: the dual problem

Definition 6: dual problem of the LP relaxation

*The **dual problem** D(LP) of the LP relaxation is the following LP problem, with optimal value $z(\mathrm{D}(\mathrm{LP}))$:*

$$\min \quad \sum_{i=1}^{m} b_i \, \pi_i$$

$$\text{subject to} \quad \sum_{i=1}^{m} a_{ij} \, \pi_i \geq c_j, \quad \forall j \in \{1, 2, \ldots, n\},$$

$$\pi_i \geq 0, \quad \forall i \in \{1, 2, \ldots, m\}.$$

Why non-negative multipliers?

- Multiplying the constraint i by $\pi_i \geq 0$ and summing, we get an inequality whose coefficients are at least the costs, by the constraints $\sum_{i=1}^{m} a_{ij}\, \pi_i \geq c_j$ of the dual problem
- The multipliers must be **non-negative**: a negative multiplier would reverse the inequality

Proposition 3: dual bounds

For every feasible solution $\bar{\pi}$ of the dual problem,

$$z(\mathrm{ILP}) \;\leq\; z(\mathrm{LP}) \;\leq\; \sum_{i=1}^{m} b_i\, \bar{\pi}_i.$$

By strong duality, $z(\mathrm{D}(\mathrm{LP})) = z(\mathrm{LP})$: the best multipliers give exactly the value of the LP relaxation.

Proof

- Let $\boldsymbol{x}$ be feasible for the LP relaxation
- Multiplying the constraint i by $\bar{\pi}_i \geq 0$ and summing:

$$\sum_{j=1}^{n} \left(\sum_{i=1}^{m} a_{ij}\, \bar{\pi}_i \right) x_j \leq \sum_{i=1}^{m} b_i\, \bar{\pi}_i$$

- By $\sum_{i=1}^{m} a_{ij}\, \bar{\pi}_i \geq c_j$ and $x_j \geq 0$, the left-hand side is at least $\sum_{j=1}^{n} c_j\, x_j$
- Hence, for every feasible $\boldsymbol{x}$,

$$\sum_{j=1}^{n} c_j\, x_j \leq \sum_{i=1}^{m} b_i\, \bar{\pi}_i$$



Dual solutions as certificates

Every feasible solution of the dual problem
is a **certificate** of an upper bound,
found without solving the LP relaxation.

The best certificate is an optimal dual solution,
and it has the same value as the LP relaxation.

Example: multipliers for the car example (1/5)

- A feasible solution of the LP relaxation does **not** give a dual bound: its value is neither a lower bound nor an upper bound on $z(\mathrm{ILP})$

(x_1, x_2)	feasible for the LP relaxation	integer	$3\,x_1 + 2\,x_2$
$(4, \frac{5}{4})$	yes	no	$\frac{29}{2} > 14$
$(2, \frac{5}{2})$	yes	no	$11 < 14$

Example: multipliers for the car example (2/5)

- Multiplying the first constraint of the car example by $\frac{3}{4}$ we get

$$\frac{15}{2} x_1 + 3 x_2 \leq \frac{135}{4}$$

- Since $x_1, x_2 \geq 0$ and the coefficients $\frac{15}{2}$ and 3 are at least the costs 3 and 2, we have

$$3 x_1 + 2 x_2 \leq \frac{15}{2} x_1 + 3 x_2 \leq \frac{135}{4}$$

- This is an upper bound of $\frac{135}{4}$

Example: multipliers for the car example (3/5)

- With multipliers $\pi_1, \pi_2 \geq 0$ of the two constraints, the sum of the constraints is

$$(10\,\pi_1 + 4\,\pi_2)\,x_1 + (4\,\pi_1 + 14\,\pi_2)\,x_2 \leq 45\,\pi_1 + 49\,\pi_2$$

- The coefficients of the sum must be at least the costs:

$$10\,\pi_1 + 4\,\pi_2 \geq 3, \qquad 4\,\pi_1 + 14\,\pi_2 \geq 2$$

- Then $45\,\pi_1 + 49\,\pi_2$ is an upper bound

Example: multipliers for the car example (4/5)

- Some multipliers and the bounds they give:

(π_1, π_2)	valid multipliers	upper bound $45\,\pi_1 + 49\,\pi_2$
$(\frac{3}{4}, 0)$	yes	$\frac{135}{4}$
$(0, \frac{3}{4})$	yes	$\frac{147}{4}$
$(\frac{1}{5}, \frac{1}{4})$	yes	$\frac{85}{4}$
$(\frac{1}{2}, 0)$	yes	$\frac{45}{2}$
$(\frac{1}{5}, 0)$	no: $10 \cdot \frac{1}{5} = 2 < 3$	–

Example: multipliers for the car example (5/5)

- The best multipliers, with the smallest value $45\,\pi_1 + 49\,\pi_2$, solve the dual problem of the car example:

$$\begin{array}{rrcrcl} \min & 45\,\pi_1 & + & 49\,\pi_2 & & \\ \text{subject to} & 10\,\pi_1 & + & 4\,\pi_2 & \geq & 3 \\ & 4\,\pi_1 & + & 14\,\pi_2 & \geq & 2 \\ & \pi_1, & & \pi_2 & \geq & 0 \end{array}$$

Example: optimal dual solution of the car example (1/2)

- The optimal solution of the dual problem of the car example is

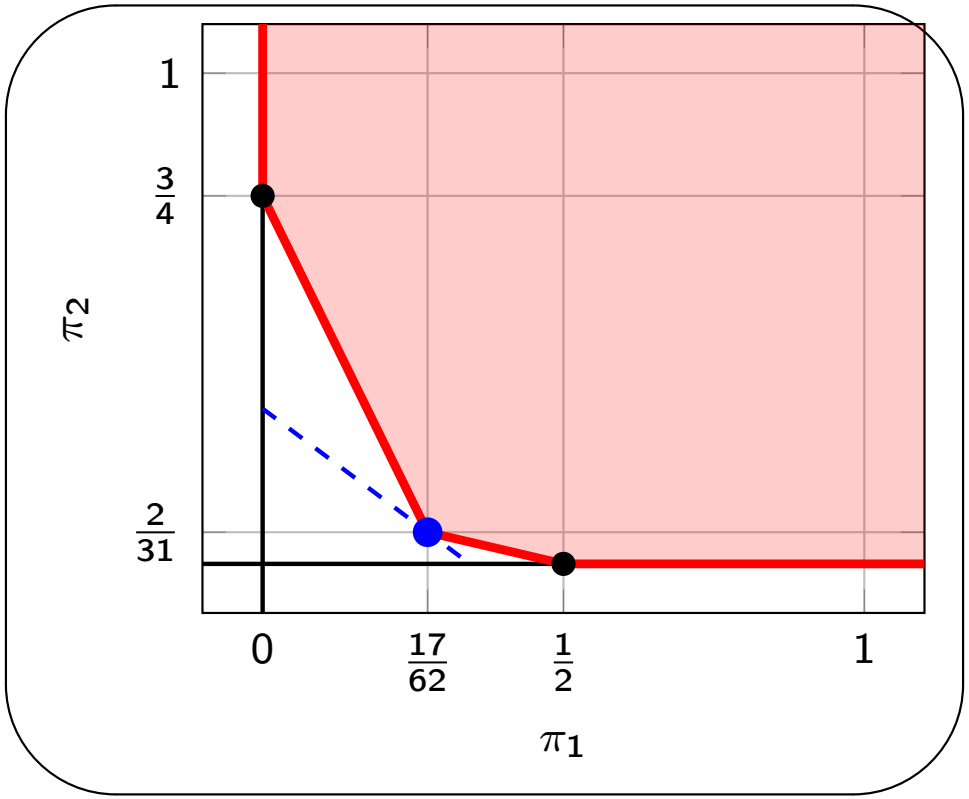
$$\tilde{\boldsymbol{\pi}}(\mathrm{D}(\mathrm{LP})) = \left(\frac{17}{62}, \frac{2}{31}\right)$$

- Its value is

$$z(\mathrm{D}(\mathrm{LP})) = 45 \cdot \frac{17}{62} + 49 \cdot \frac{2}{31} = \frac{765}{62} + \frac{98}{31} = \frac{31}{2} = z(\mathrm{LP})$$

- The other two vertices of the dual feasible region give the bounds $\frac{45}{2}$ and $\frac{147}{4}$

Example: optimal dual solution of the car example (2/2)



The feasible region of the dual LP problem (red, unbounded) and its optimal solution $(\frac{17}{62}, \frac{2}{31})$ (blue), with the objective line $45\,\pi_1 + 49\,\pi_2 = \frac{31}{2}$ (dashed). The other vertices (black) give the bounds $\frac{45}{2}$ and $\frac{147}{4}$.

Summary of the notation

- **ILP problem**: optimal solution $\tilde{\boldsymbol{x}}(\mathrm{ILP})$, optimal value $z(\mathrm{ILP})$
- **Primal bound**: feasible solution $\bar{\boldsymbol{x}}$ with $\mathrm{LB}(\bar{\boldsymbol{x}}) = \sum_j c_j \bar{x}_j \leq z(\mathrm{ILP})$
- **LP relaxation**: optimal solution $\tilde{\boldsymbol{x}}(\mathrm{LP})$, optimal value $z(\mathrm{LP})$, upper bound $z(\mathrm{ILP}) \leq z(\mathrm{LP})$
- **Dual problem**: optimal solution $\tilde{\boldsymbol{\pi}}(\mathrm{D}(\mathrm{LP}))$, optimal value $z(\mathrm{D}(\mathrm{LP})) = z(\mathrm{LP})$, and $z(\mathrm{ILP}) \leq \sum_i b_i \bar{\pi}_i$ for every dual feasible $\bar{\boldsymbol{\pi}}$

Summary

- Non-negative multipliers π_i of the constraints, with combined coefficients at least the costs, give the dual bound $\sum_i b_i \pi_i$
- The best multipliers solve the **dual problem** D(LP), and by strong duality $z(\mathrm{D}(\mathrm{LP})) = z(\mathrm{LP})$
- The value of a feasible LP solution is not a bound
- Car example: $\tilde{\boldsymbol{\pi}}(\mathrm{D}(\mathrm{LP})) = (\frac{17}{62}, \frac{2}{31})$, with value $\frac{31}{2} = z(\mathrm{LP})$

Typical mistakes

Introduction: Typical mistakes

- We list the mistakes that are made most often with primal and dual bounds
- They concern certificates, rounding, multipliers and bounds on the variables
- Knowing them helps to read and solve the exercises correctly

Typical mistakes

Observation 2: Typical mistakes (1/3)

- ***Taking a heuristic solution as optimal.*** *A feasible solution gives only a primal bound. Without a dual bound with the same value there is no certificate of optimality.*
- ***Rounding the bound when it is not valid.*** *The bound* $\lfloor z(\mathrm{LP}) \rfloor$ *requires all the variables and all the costs to be integer. In a MIP problem the bound is* $z(\mathrm{LP})$.

Typical mistakes

Observation: Typical mistakes (2/3)

- ***Rounding the solution and calling it optimal.*** *A rounded solution is feasible at best, and in general not optimal. It may even be infeasible, as the nearest integer point* $(2, 5)$ *of the example «rounding fails».*
- ***Using the value of a feasible LP solution as a bound.*** *Only the optimal value of the LP relaxation, or the value of a dual feasible solution, is an upper bound.*

Typical mistakes

Observation: Typical mistakes (3/3)

- ***Multipliers with a negative sign.*** *The multipliers of constraints $\leq$ must be non-negative, and the combined coefficients must be at least the costs.*
- ***Deriving a bound on a variable from a constraint with negative coefficients.*** *The proof of $x_j \leq \lfloor b_i / a_{ij} \rfloor$ needs all the coefficients to be non-negative.*
- ***Believing that a stronger formulation always improves the bound.*** *The bounds $x_1 \leq 4$ and $x_2 \leq 3$ leave the bound 15 of the car example unchanged.*

Summary

Typical mistakes:

- A heuristic solution is not a certificate; a rounded solution is feasible at best
- $\lfloor z(\mathrm{LP}) \rfloor$ needs integer variables and integer costs
- The value of a feasible LP solution is not a bound
- Multipliers of $\leq$ constraints are non-negative; bounds on the variables need non-negative coefficients
- A stronger formulation does not always improve the bound

In summary

Introduction: In summary

- We collect the key ideas of the chapter
- We close with the dashboard of the car example: the bounds found so far

Summary (1/3)

- Every feasible solution is a primal bound (a lower bound) on $z(\mathrm{ILP})$, and heuristic algorithms find good feasible solutions quickly
- The LP relaxation drops integrality, so $z(\mathrm{ILP}) \leq z(\mathrm{LP})$; with integer variables and integer costs this dual bound (an upper bound) can be rounded down to $\lfloor z(\mathrm{LP}) \rfloor$
- Primal and dual bound define the optimality gap; optimality is certified when the gap is zero

Summary (2/3)

- Rounding an optimal LP solution gives a feasible solution at best, never a certificate
- The upper bounds on the variables $x_j \leq \lfloor b_i / a_{ij} \rfloor$ come from constraints with non-negative coefficients, and they help only if they cut off the LP optimum
- Non-negative multipliers of the constraints give dual bounds; the best ones solve the dual problem, whose optimal value is $z(\mathrm{LP})$

Summary (3/3)

Dashboard of the car example. The bounds on $z(\mathrm{ILP})$ found so far in this module (primal bound: lower; dual bound: upper):

Chapter	Lower bound	Upper bound	From
Primal and dual bounds	14	15	heuristic solution $(4, 1)$, and $\lfloor z(\mathrm{LP}) \rfloor$ with $z(\mathrm{LP}) = \frac{31}{2}$